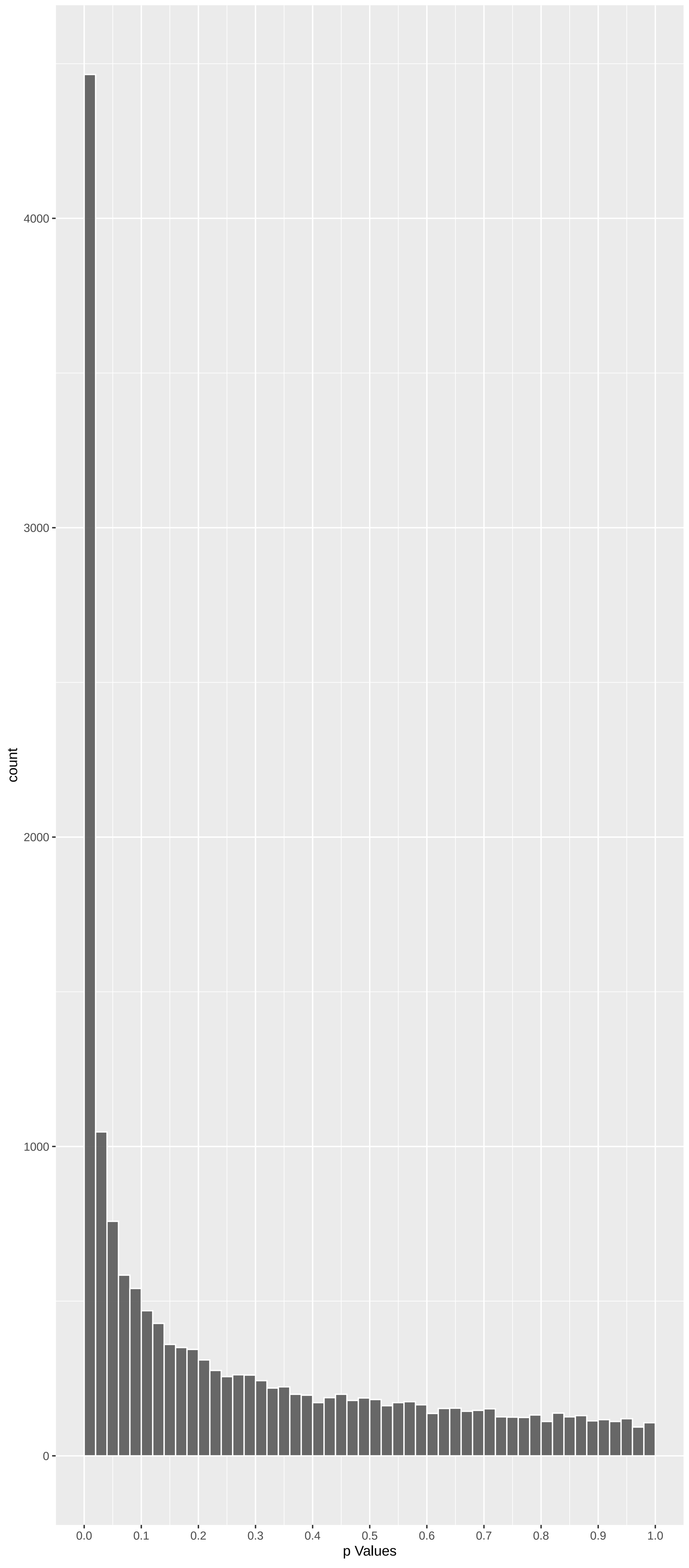
Top genes by P−value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
SPG11	0.09921964	627	3.256e−14	5.252e−10	2.040e−06
TRPM5	0.11837287	511	7.777e−14	6.273e−10	2.040e−06
EP400	0.10360299	572	1.426e−13	7.669e−10	2.040e−06
ZFAND4	0.11180013	517	2.972e−13	1.199e−09	2.391e−06
TRIM15	0.13087578	423	1.016e−12	3.279e−09	3.432e−06
PLEC	0.11871026	461	1.388e−12	3.731e−09	3.432e−06
ADGRF1	0.08909133	604	2.160e−12	3.871e−09	3.432e−06
ECI2	0.10166850	530	2.095e−12	3.871e−09	3.432e−06
TMEM131L	0.09193039	590	1.746e−12	3.871e−09	3.432e−06
SLC14A2	0.09200406	573	3.735e−12	6.026e−09	4.421e−06
PJA2	0.10292213	506	5.172e−12	7.584e−09	4.421e−06
CLCNKA	0.10289648	501	6.732e−12	7.757e−09	4.421e−06
AKAP9	0.08725110	592	6.349e−12	7.757e−09	4.421e−06
DNAH14	0.10140921	509	6.508e−12	7.757e−09	4.421e−06
MTBP	0.08607087	588	1.069e−11	1.150e−08	6.115e−06
MMEL1	0.09778120	502	2.303e−11	2.322e−08	1.060e−05
AGGF1	0.09764501	501	2.502e−11	2.374e−08	1.060e−05
CLCNKB	0.10236078	475	2.912e−11	2.524e−08	1.060e−05
SPEGNB	0.21077320	231	2.973e−11	2.524e−08	1.060e−05
RIMBP2	0.08794335	541	4.873e−11	3.931e−08	1.568e−05
CLDN16	0.12205728	382	7.977e−11	6.057e−08	2.101e−05
CFAP20DC	0.08305575	559	8.636e−11	6.057e−08	2.101e−05
FGD5	0.08966837	518	8.578e−11	6.057e−08	2.101e−05
TACR2	0.10287762	446	1.145e−10	7.698e−08	2.559e−05
PPP4R1	0.08781879	516	1.511e−10	9.748e−08	3.111e−05

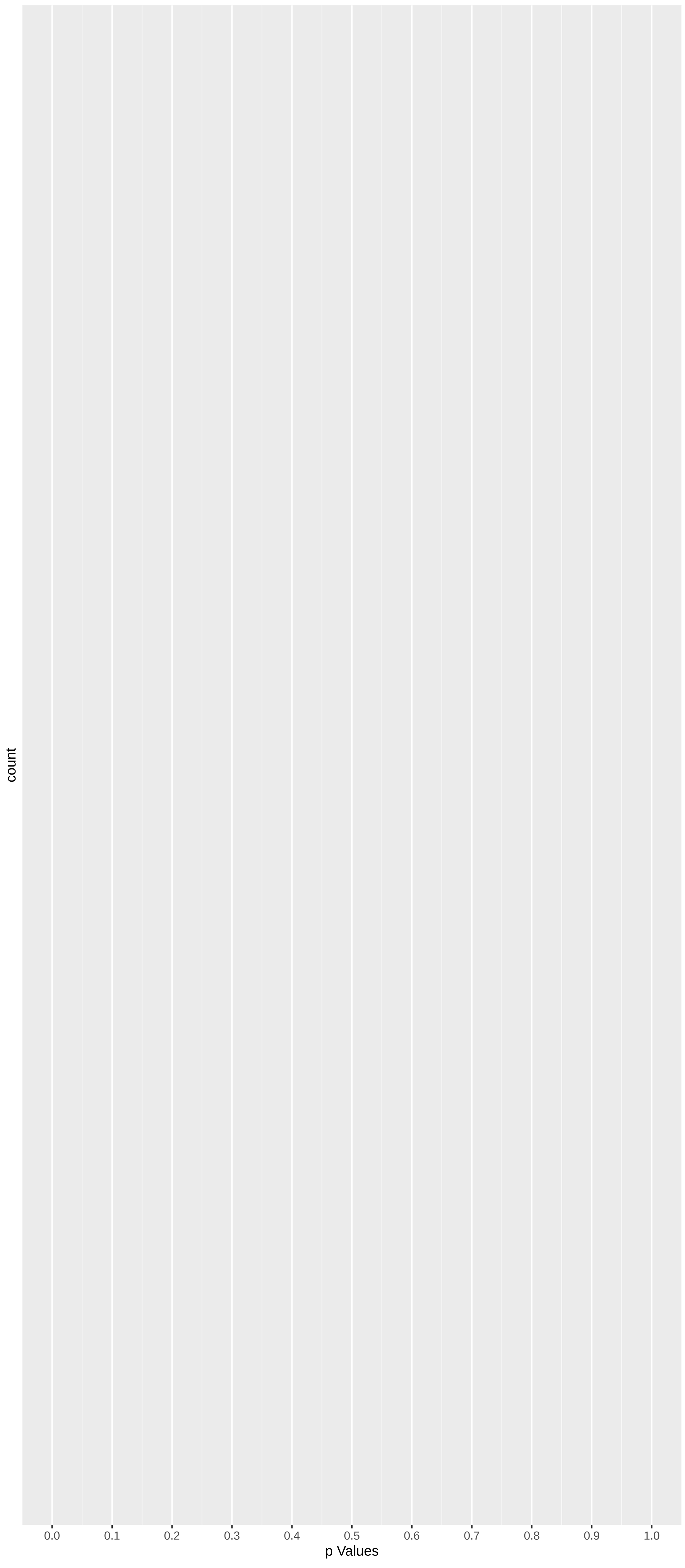
Top genes by Q−Value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
SPG11	0.09921964	627	3.256e−14	5.252e−10	2.040e−06
EP400	0.10360299	572	1.426e−13	7.669e−10	2.040e−06
TRPM5	0.11837287	511	7.777e−14	6.273e−10	2.040e−06
ZFAND4	0.11180013	517	2.972e−13	1.199e−09	2.391e−06
PLEC	0.11871026	461	1.388e−12	3.731e−09	3.432e−06
ADGRF1	0.08909133	604	2.160e−12	3.871e−09	3.432e−06
TRIM15	0.13087578	423	1.016e−12	3.279e−09	3.432e−06
ECI2	0.10166850	530	2.095e−12	3.871e−09	3.432e−06
TMEM131L	0.09193039	590	1.746e−12	3.871e−09	3.432e−06
CLCNKA	0.10289648	501	6.732e−12	7.757e−09	4.421e−06
AKAP9	0.08725110	592	6.349e−12	7.757e−09	4.421e−06
PJA2	0.10292213	506	5.172e−12	7.584e−09	4.421e−06
DNAH14	0.10140921	509	6.508e−12	7.757e−09	4.421e−06
SLC14A2	0.09200406	573	3.735e−12	6.026e−09	4.421e−06
MTBP	0.08607087	588	1.069e−11	1.150e−08	6.115e−06
AGGF1	0.09764501	501	2.502e−11	2.374e−08	1.060e−05
CLCNKB	0.10236078	475	2.912e−11	2.524e−08	1.060e−05
MMEL1	0.09778120	502	2.303e−11	2.322e−08	1.060e−05
SPEGNB	0.21077320	231	2.973e−11	2.524e−08	1.060e−05
RIMBP2	0.08794335	541	4.873e−11	3.931e−08	1.568e−05
CLDN16	0.12205728	382	7.977e−11	6.057e−08	2.101e−05
CFAP20DC	0.08305575	559	8.636e−11	6.057e−08	2.101e−05
FGD5	0.08966837	518	8.578e−11	6.057e−08	2.101e−05
TACR2	0.10287762	446	1.145e−10	7.698e−08	2.559e−05
PPP4R1	0.08781879	516	1.511e−10	9.748e−08	3.111e−05

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
SPG11	0.09921964	627	3.256e−14	5.252e−10	2.040e−06
TRPM5	0.11837287	511	7.777e−14	6.273e−10	2.040e−06
EP400	0.10360299	572	1.426e−13	7.669e−10	2.040e−06
ZFAND4	0.11180013	517	2.972e−13	1.199e−09	2.391e−06
TRIM15	0.13087578	423	1.016e−12	3.279e−09	3.432e−06
PLEC	0.11871026	461	1.388e−12	3.731e−09	3.432e−06
ADGRF1	0.08909133	604	2.160e−12	3.871e−09	3.432e−06
ECI2	0.10166850	530	2.095e−12	3.871e−09	3.432e−06
TMEM131L	0.09193039	590	1.746e−12	3.871e−09	3.432e−06
SLC14A2	0.09200406	573	3.735e−12	6.026e−09	4.421e−06

Top Negative genes by P−value non−permulated

Gene	Rho	N	P	p.adj	qValueNoperm
NA	NA	NA	NA	NA	NA
NA.1	NA	NA	NA	NA	NA
NA.2	NA	NA	NA	NA	NA
NA.3	NA	NA	NA	NA	NA
NA.4	NA	NA	NA	NA	NA
NA.5	NA	NA	NA	NA	NA
NA.6	NA	NA	NA	NA	NA
NA.7	NA	NA	NA	NA	NA
NA.8	NA	NA	NA	NA	NA
NA.9	NA	NA	NA	NA	NA

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HOUNKPE_HOUSEKEEPING_GENES	−0.07346046	1061	1.198e−15	7.773e−12	STK38:8 CCT2:23 DERL1:24 CMPK1:34 HAX1:39 EIF2S1:47
REACTOME_METABOLISM_OF_RNA	−0.08325670	641	8.579e−13	2.783e−09	SF3B3:15 RAN:16 PSMD8:73 APOBEC1:85 GEMIN4:101 NCBP2:106
BLALOCK_ALZHEIMERS_DISEASE_DN	−0.06132198	1150	4.049e−12	8.756e−09	RSU1:6 LAPTM4B:9 CCT2:23 MAPK9:37 CDK14:44 EIF2S1:47
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	−0.07732462	692	5.598e−12	9.079e−09	NLRP3:2 CCS:27 DEDD2:35 MAPK9:37 EIF2S1:47 LMNB1:52
REACTOME_REGULATION_OF_MRNA_STABILITY_BY	−0.20730775	83	6.798e−11	8.821e−08	PSMD8:73 ELAVL1:142 PSMB5:223 ANP32A:261 HSPA8:292 PSMA1:344
REACTOME_THE_ROLE_OF_GTSE1_IN_G2_M_PROGR	−0.21648249	73	1.635e−10	1.768e−07	PSMD8:73 MAPRE1:134 PSMB5:223 TUBB2B:299 PSMA1:344 PSMC3:451
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	−0.05976798	1011	1.913e−10	1.773e−07	TRIAP1:33 HES1:41 YEATS4:61 GAMT:64 CDC27:68 NBN:69
REACTOME_INFECTIOUS_DISEASE	−0.06601246	809	2.378e−10	1.929e−07	NLRP3:2 RAN:16 TAB2:19 PSMD8:73 GEMIN4:101 NCBP2:106
REACTOME_REGULATION_OF_EXPRESSION_OF_SLI	−0.14962606	146	4.559e−10	3.286e−07	PSMD8:73 NCBP2:106 RPS13:196 PSMB5:223 ELOB:231 MAGOHB:282
KIM_BIPOLAR_DISORDER_OLIGODENDROCYTE_DEN	−0.07076215	624	1.967e−09	1.276e−06	NSF:7 RAN:16 CLTB:18 CCT2:23 MAPK9:37 PPP6C:55
REACTOME_SIGNALING_BY_ROBO_RECEPTORS	−0.12426449	189	4.051e−09	2.389e−06	PSMD8:73 NCBP2:106 RPS13:196 PSMB5:223 ELOB:231 MAGOHB:282
DIAZ_CHRONIC_MYELOGENOUS_LEUKEMIA_UP	−0.04746257	1319	1.094e−08	5.913e−06	RSU1:6 LAPTM4B:9 SF3B3:15 CCT2:23 CMPK1:34 KCTD3:42
REACTOME_SIGNALING_BY_THE_B_CELL_RECEPTO	−0.15819368	105	2.189e−08	1.063e−05	PSMD8:73 BLK:147 ORAI1:217 PSMB5:223 PSMA1:344 SH3KBP1:422
PUJANA_BRCA1_PCC_NETWORK	−0.04406881	1479	2.294e−08	1.063e−05	ITPK1:11 RAN:16 CCT2:23 TRIAP1:33 CHST7:43 EIF2S1:47
MILI_PSEUDOPODIA_HAPTOTAXIS_UP	−0.07403883	485	2.690e−08	1.141e−05	PDCD10:48 MRPS18C:71 STYX:128 PTAR1:143 TIMM8A:195 SRP54:197
REACTOME_CLEC7A_DECTIN_1_SIGNALING	−0.16582669	94	2.815e−08	1.141e−05	TAB2:19 PSMD8:73 PSMB5:223 CLEC7A:320 PSMA1:344 PSMC3:451
REACTOME_UCH_PROTEINASES	−0.17278374	85	3.730e−08	1.407e−05	PSMD8:73 TFPT:129 PSMB5:223 PSMA1:344 PSMC3:451 USP15:517
BORCZUK_MALIGNANT_MESOTHELIOMA_UP	−0.09434338	288	3.905e−08	1.407e−05	RAN:16 CCT2:23 TRIAP1:33 MAPK9:37 CDC27:68 PSMD8:73
ACEVEDO_METHYLATED_IN_LIVER_CANCER_DN	0.06362738	637	5.063e−08	1.729e−05	FGD5:22 ADGRV1:31 ADGRG7:34 ANK1:42 TIAM2:82 SLC43A1:96
REACTOME_MRNA_SPLICING	−0.11347883	193	5.718e−08	1.855e−05	SF3B3:15 NCBP2:106 SNRPB2:249 MAGOHB:282 HSPA8:292 LSM4:360
REACTOME_HIV_INFECTION	−0.10597977	214	9.619e−08	2.837e−05	RAN:16 PSMD8:73 NCBP2:106 PSMB5:223 ELOB:231 CD8B:239
HSIAO_HOUSEKEEPING_GENES	−0.08545768	332	9.467e−08	2.837e−05	ITPK1:11 PSMD8:73 TMED10:183 PSMB5:223 COX4I1:287 RPLP0:328
REACTOME_FCERI_MEDIATED_NF_KB_ACTIVATION	−0.17521528	77	1.079e−07	3.044e−05	TAB2:19 PSMD8:73 PSMB5:223 FCER1G:274 PSMA1:344 PSMC3:451
REACTOME_RUNX1_REGULATES_TRANSCRIPTION_O	−0.16348035	88	1.174e−07	3.175e−05	PSMD8:73 PSMB5:223 PSMA1:344 CDK7:361 H4C13:393 PSMC3:451
PUJANA_CHEK2_PCC_NETWORK	−0.05850112	704	1.478e−07	3.836e−05	RAN:16 CCT2:23 TRIAP1:33 EIF2S1:47 PDCD10:48 LMNB1:52
REACTOME_SCF_BETA_TRCP_MEDIATED_DEGRADAT	−0.20989914	52	1.655e−07	4.003e−05	PSMD8:73 CDC20:111 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566
REACTOME_DNA_REPLICATION_PRE_INITIATION	−0.14200887	114	1.666e−07	4.003e−05	CDC27:68 PSMD8:73 PSMB5:223 PSMA1:344 H4C13:393 PSMC3:451
REACTOME_DOWNSTREAM_SIGNALING_EVENTS_OF_	−0.17345803	76	1.737e−07	4.025e−05	PSMD8:73 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566 NFKBIA:799
WANG_TUMOR_INVASIVENESS_UP	−0.08111609	351	1.937e−07	4.201e−05	CCT2:23 STX7:72 NHP2:121 ELAVL1:142 SCAF8:145 COX4I1:287
REACTOME_C_TYPE_LECTIN_RECEPTORS_CLRS	−0.13942790	117	1.943e−07	4.201e−05	TAB2:19 PSMD8:73 PSMB5:223 FCER1G:274 CLEC7A:320 PSMA1:344
REACTOME_INNATE_IMMUNE_SYSTEM	−0.05235453	865	2.157e−07	4.488e−05	NLRP3:2 TXK:3 TAB2:19 CCT2:23 MAPK9:37 VPS35L:40
REACTOME_APC_C_MEDIATED_DEGRADATION_OF_C	−0.16458800	83	2.214e−07	4.488e−05	CDC27:68 PSMD8:73 CDC20:111 PSMB5:223 PSMA1:344 PSMC3:451
REACTOME_DECTIN_1_MEDIATED_NONCANONICAL_	−0.19534626	58	2.691e−07	5.146e−05	PSMD8:73 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566 PSMA8:801
KEGG_SPLICEOSOME	−0.14271748	109	2.697e−07	5.146e−05	SF3B3:15 NCBP2:106 SNRPB2:249 MAGOHB:282 HSPA8:292 LSM4:360
REACTOME_TCR_SIGNALING	−0.14409078	106	3.031e−07	5.314e−05	TAB2:19 PSMD8:73 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566
REACTOME_DISEASES_OF_SIGNAL_TRANSDUCTION	−0.07359314	415	2.986e−07	5.314e−05	HES1:41 PSMD8:73 NCBP2:106 CNKSR2:118 KREMEN1:131 NR4A1:174
BENPORATH_MYC_MAX_TARGETS	−0.05756673	691	2.966e−07	5.314e−05	MST1R:1 HAX1:39 EIF2S1:47 PDCD10:48 PPP6C:55 STX7:72
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	−0.10892158	186	3.133e−07	5.349e−05	PSMD8:73 BLK:147 PSMB5:223 PSMA1:344 CDK7:361 H4C13:393
REACTOME_MITOTIC_G1_PHASE_AND_G1_S_TRANS	−0.12342766	144	3.271e−07	5.442e−05	PSMD8:73 PSMB5:223 PSMA1:344 CDK7:361 PSMC3:451 MYBL2:553
MARTENS_TRETINOIN_RESPONSE_DN	−0.05691750	699	3.475e−07	5.636e−05	RAN:16 TRIAP1:33 HES1:41 MRPS18C:71 PEX12:89 GEMIN4:101

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
22q11 Deletion Syndrome	0.28414095	19	1.810e−05	5.919e−02	HIRA:85 JMJD1C:603 CHRD:671 ZDHHC8:1370 DGCR8:1503 COMT:1536
Glioma	−0.03194195	1776	1.495e−05	5.919e−02	MST1R:1 RSU1:6 BCL2L12:32 CMPK1:34 MAPK9:37 HES1:41
Tetany	0.28378849	21	6.759e−06	5.919e−02	CLDN16:21 HIRA:85 TRPM6:103 JMJD1C:603 KCNJ1:644 SLC12A3:767
Glioblastoma	−0.03110326	1593	5.686e−05	1.116e−01	RSU1:6 BCL2L12:32 MAPK9:37 HES1:41 PDCD10:48 RARRES2:57
Precursor T−Cell Lymphoblastic Leukemia−	−0.06466747	337	4.949e−05	1.116e−01	HES1:41 ZEB2:86 IGFBP7:119 TIMM8A:195 NKX2−5:221 MSX2:226
Neoplasm Metastasis	−0.02359383	3131	6.932e−05	1.133e−01	MST1R:1 RSU1:6 LAPTM4B:9 RAN:16 BRMS1L:17 COL6A3:21
Anemia, hereditary spherocytic hemolytic	0.48861271	5	1.544e−04	1.515e−01	ANK1:42 SPTA1:70 SLC4A1:130 EPB42:275 SPTB:394 NA
Carcinogenesis	−0.02209373	3287	1.527e−04	1.515e−01	MST1R:1 NLRP3:2 RSU1:6 STK38:8 LAPTM4B:9 RAN:16
Classical Hodgkin's Lymphoma	−0.09338934	142	1.263e−04	1.515e−01	MST1R:1 FSCN1:210 CYSLTR1:238 TNFRSF17:315 EPHB2:454 IL3RA:567
Spherocytosis	0.48861271	5	1.544e−04	1.515e−01	ANK1:42 SPTA1:70 SLC4A1:130 EPB42:275 SPTB:394 NA
Cytochrome−c Oxidase Deficiency	−0.17963093	36	1.929e−04	1.582e−01	CCS:27 COA7:136 COA6:159 COX4I1:287 ETHE1:592 COX5A:973
Lymphoma	−0.03432349	1072	1.936e−04	1.582e−01	MST1R:1 NLRP3:2 NSF:7 STK38:8 NBN:69 TNFSF13B:139
Malignant neoplasm of stomach	−0.02652887	1876	2.342e−04	1.767e−01	MST1R:1 STK38:8 LAPTM4B:9 COL6A3:21 BCL2L12:32 HES1:41
Central neuroblastoma	−0.02918871	1407	3.443e−04	2.252e−01	NSF:7 RAN:16 MAPK9:37 HES1:41 IRF2:50 IL31RA:78
Squamous cell carcinoma	−0.02843021	1500	3.369e−04	2.252e−01	MST1R:1 MAPK9:37 HAX1:39 HES1:41 CDK14:44 EIF2S1:47
HIV Infections	−0.04054014	634	5.652e−04	3.465e−01	RAN:16 CMPK1:34 MAPK9:37 NBN:69 PSMD8:73 CCL19:179
22q11 partial monosomy syndrome	0.29166078	11	8.100e−04	3.484e−01	HIRA:85 JMJD1C:603 COMT:1536 ARVCF:1658 PI4KA:2297 TBX1:2862
Bilateral cataracts (disorder)	0.09504775	103	8.745e−04	3.484e−01	GSTO1:117 CRYGA:378 SMN2:499 BFSP2:616 NFE2L2:664 SMN1:676
Broad chest	−0.35930702	7	9.945e−04	3.484e−01	BRAF:526 MAP2K1:675 PTPN11:763 CHST3:2186 DYM:3635 TRAPPC2:3870
Familial intrahepatic cholestasis of pre	0.39443020	6	8.200e−04	3.484e−01	ATP8B1:334 ABCB4:1161 NR1H4:1321 VCAM1:1624 ABCB11:1746 TNF:3765
Hamartoma of tongue	0.42532236	5	9.885e−04	3.484e−01	OFD1:692 C2CD3:765 WDPCP:1022 TCTN3:1434 NEK1:1923 NA
Leukemogenesis	−0.03994559	617	7.974e−04	3.484e−01	HAX1:39 HES1:41 NR4A1:174 NKX2−5:221 CD8B:239 PRKAA2:255
Metabolic Bone Disorder	0.17754689	30	7.664e−04	3.484e−01	CA2:459 CYP2R1:482 PTH:1027 ESR1:1070 IL6:1085 MMP2:1182
Neuroblastoma	−0.02659929	1445	9.663e−04	3.484e−01	NSF:7 RAN:16 CMPK1:34 MAPK9:37 HES1:41 IRF2:50
Nuclear cataract	0.17341354	32	6.892e−04	3.484e−01	ERCC6:489 CRYAA:942 TGM2:1058 GJA8:1109 GJA3:1115 HSF4:1170
Posterior subcapsular cataract	0.22135469	20	6.116e−04	3.484e−01	NFE2L2:664 ZNF408:813 GJA8:1109 GJA3:1115 ADAM9:1142 HSF4:1170
Shield chest	−0.35930702	7	9.945e−04	3.484e−01	BRAF:526 MAP2K1:675 PTPN11:763 CHST3:2186 DYM:3635 TRAPPC2:3870
Stomach Carcinoma	−0.02457228	1837	7.302e−04	3.484e−01	MST1R:1 STK38:8 LAPTM4B:9 COL6A3:21 BCL2L12:32 HES1:41
Mammary Neoplasms	−0.02299701	1989	1.087e−03	3.678e−01	MST1R:1 BRMS1L:17 DERL1:24 BCL2L12:32 MAPK9:37 HES1:41
Ciliopathies	0.07255673	170	1.133e−03	3.704e−01	FAM161A:218 ARL13B:342 CRB2:352 INVS:355 MAPKBP1:418 CCDC40:504
CATARACT, COPPOCK−LIKE	0.28873261	10	1.569e−03	3.723e−01	ISG20:384 BFSP2:616 GJA8:1109 GJA3:1115 CRYBB2:2433 CRYBB1:2535
Cataract, Pulverulent	0.28368172	10	1.895e−03	3.723e−01	BFSP2:616 GJA8:1109 GJA3:1115 HSF4:1170 CRYBB2:2433 CRYBB1:2535
Leukemia, Myelocytic, Acute	−0.02617473	1376	1.480e−03	3.723e−01	DERL1:24 CMPK1:34 HES1:41 IRF2:50 NBN:69 ZEB2:86
Night blindness, congenital stationary	0.17773767	25	2.102e−03	3.723e−01	GPR179:109 PDE6B:262 USP11:604 TRPM1:629 GPR34:1366 RBP3:1828
Polyarticular Juvenile Idiopathic Arthri	−0.21031283	18	2.011e−03	3.723e−01	IL2RB:372 PRR5L:636 IL2RA:1618 IL6R:1966 STAT1:2316 STAT4:2395
Syndactyly, type 2	−0.31862326	8	1.804e−03	3.723e−01	HOXD8:269 CD6:308 HOXA13:1038 HOXD9:1798 FBLN1:3563 RASSF8:4340
Abnormality of macular pigmentation	0.19348144	22	1.684e−03	3.723e−01	GPR179:109 PDE6B:262 PROM1:567 TRPM1:629 RD3:720 ELOVL4:1744
Acute kidney injury	0.10745317	72	1.635e−03	3.723e−01	ALB:69 HPX:223 LPIN1:226 IGFBP1:285 HEXB:369 C3:643
Adult T−Cell Lymphoma/Leukemia	−0.04449278	410	2.136e−03	3.723e−01	NLRP3:2 GEMIN4:101 CCL19:179 FSCN1:210 STAMBPL1:222 IRF3:228
Aplasia/Hypoplasia of the cerebellar ver	0.20296308	19	2.196e−03	3.723e−01	CRB1:48 SPECC1L:484 RD3:720 CEP290:846 KCNJ13:952 RDH12:1192